<sol_start id=1>
1. (3) 3
<sol_end>

<sol_start id=2>
2. (2) Sin60°
<sol_end>

<sol_start id=3>
3. (4) 100
<sol_end>

<sol_start id=4>
4 (1) 10 cm
<sol_end>

<sol_start id=5>
5 (4) a = 0
<sol_end>

<sol_start id=6>
6. (4) Four decimal places
<sol_end>

<sol_start id=7>
7. (3) 3 and 1
<sol_end>

<sol_start id=8>
8. (2) (0, –1)
<sol_end>

<sol_start id=9>
9. (4) No solution
<sol_end>

<sol_start id=10>

10. (1) $\frac{-9}{2}$
<sol_end>

<sol_start id=11>
11. (1) $k \neq 3$
<sol_end>

<sol_start id=12>
12. (1) 8
<sol_end>

<sol_start id=13>
13. (1) similar
<sol_end>

<sol_start id=14>
14. (1) $x^3y^4z^4$
<sol_end>

<sol_start id=15>
15. (3) $\frac{x^2}{2}-\frac{x}{2}-6$
<sol_end>

<sol_start id=16>
16. (3) Intersecting or coincident
<sol_end>

<sol_start id=17>
17. (1) Parallel
<sol_end>

<sol_start id=18>
18. (4) 16 : 81
<sol_end>

<sol_start id=19>

19. (3) Assertion (A) is true but reason (R) is false

<sol_end>

<sol_start id=20>

20. (1) Both A and R are true and R is the correct explanation of A

<sol_end>

<sol_start id=21>

21. → Given – a number in the form,

$6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$

$144 \times 5 + 5$

$5\ (144 + 1)$

$(5)\ (145)$

$(5)\ (5)\ (29)\ (1)$

∴ The given no. is a composite number as it has more than two factors

<sol_end>

<sol_start id=22>

22. → Given – Pair of linear equations–

$4x + 6y = ?\ ;\ kx + 18y = 21$

Solution – condition for infinitely many solutions

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\frac{4}{k} = \frac{\cancel{6}}{\cancel{18}} = \frac{\cancel{7}}{\cancel{21}}$$

$$\frac{4}{k} = \frac{1}{3}$$

$4 \times 3 = k$

$12 = k$

∴ The value of k will be 12 for linear equation pair with coincident lines.

<sol_end>

<sol_start id=23>

23. $\rightarrow$ Given – Point (−6,8) , Finding its distance from origin,

Solution – using the formula –

$= \sqrt{x^2 + y^2}$

$= \sqrt{(-6)^2 + (8)^2}$

$= \sqrt{36 + 64}$

$= \sqrt{100}$ = 10 units

$\therefore$ The distance of the point form origin is 10 units.

<sol_end>

<sol_start id=24>

24. $\rightarrow$ Given – sin3A = cos(A–10°) finding A –

Solution –

sin3A = cos(A–10°)

$\cancel{\cos}(90° - 3A) = \cancel{\cos}(A - 10°)$ ……[sin A = cos(90°–A)]A–90]

90°–3A = A – 10°

90° + 10° = 4A

A = $\frac{100°}{4}$ = 25°

A = 25°

<sol_end>

<sol_start id=25>

25. $\rightarrow$ Given – Fig. with ΔAOB and ΔDOC , AB||DC

To prove – $\Delta AOB \sim \Delta DOC$

Solution – In ΔAOB and ΔDOC,

$\angle$AOB = $\angle$DOC ……[Vertically opposite angle]

$\angle$ABO = $\angle$DCO …….[Alternate interior angle]

$\angle$BAO = $\angle$CDO …….[Alternate interior angle]

$\therefore$ $\Delta AOB \sim \Delta DOC$ …….[By AA similarity]

$\therefore$ Hence, proved

<sol_end>

<sol_start id=26>

26. To prove – $\dfrac{\cos(90°-\theta)}{1+\sin(90°-\theta)}+\dfrac{1+(\sin 90°-\theta)}{\cos(90°-\theta)}=2\cos ec$

Solution – using cos(90–θ) = sin θ and solving LHS

sin(90– θ) = cos θ

$$\frac{\cos(90°-\theta)}{1+\sin(90°-\theta)}+\frac{1+(\sin 90°-\theta)}{\cos(90°-\theta)}=2\cos ec \quad (1)$$

$= \dfrac{\sin\theta}{1+\cos\theta}+\dfrac{1+\cos\theta}{\sin\theta}$ ......[cos(90– θ) = sin θ, sin(90– θ) = cosθ]

$= \dfrac{\sin\theta\times\sin\theta}{1+\cos\theta\times\sin\theta}+\dfrac{1+\cos\theta\times 1+\cos\theta}{\sin\theta\times 1+\cos\theta}$

$= \dfrac{\sin^2\theta}{\sin\theta+\sin\cos\theta}+\dfrac{(1+\cos\theta)^2}{\sin\theta+\sin\cos\theta}$

$= \dfrac{\sin^2\theta+1+\cos^2\theta+2\cos\theta}{\sin\theta+\sin\cos\theta}$ ....[using $(a+b)^2 = a^2 + b^2 + 2ab$]

$= \dfrac{\sin^2\theta+\cos^2\theta+1+2\cos\theta}{\sin\theta(1+\cos\theta)}$

$= \dfrac{2+1+2\cos\theta}{\sin\theta(1+\cos\theta)}$ ......[$\sin^2\theta + \cos^2\theta = 1$]

$\dfrac{2+2\cos\theta}{\sin\theta(1+\cos\theta)}$

$\dfrac{2\cancel{(1+\cos\theta)}}{\sin\theta\cancel{(1+\cos\theta)}}$

$= \dfrac{2}{\sin\theta}$

2 cosec θ = 2 cosec (1) .... $\left[\dfrac{1}{\sin\theta}=\cos ec\theta\right]$

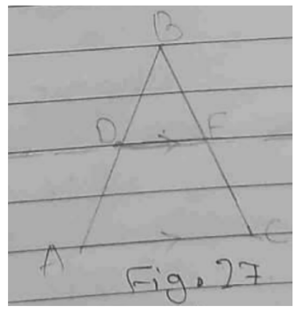
∴ LHS = RHS

∴ Hence proved

<sol_end>

<sol_start id=27>

27. Basic proportionality theorem or Thales theorem states that if a line parallel to one of the sides of the triangle when passes through the other two sides, then the other two sides will be in the same ratio.



To prove – $\frac{AD}{BD}=\frac{BE}{EC}$

Solution – Proving $\Delta ABC \sim \Delta DBC$,

In ΔABC and ΔDBE

$\angle BAC = \angle BDE$ ……[Corresponding angles]

$\angle ABC = \angle DBE$ ……[Common]

$\therefore \Delta ABC \sim \Delta DBE$ …….[By AA similiarity]

By similarity

$$\frac{AB}{DB}=\frac{CB}{EC}$$

$$\frac{AD+DE}{DB}=\frac{CE+BE}{CE}$$

$$\frac{AD}{BD}+\frac{\cancel{BD}}{\cancel{BD}}=\frac{\cancel{CE}}{\cancel{CE}}+\frac{BE}{CE}$$

$$\therefore \frac{AD}{BD}+\frac{BE}{EC}$$

$\therefore$ Hence proved.

<sol_end>

<sol_start id=28>

28. Given – Book when rented has fixed charge for two days and extra charge for each day after that,

Latika paid 22 for 6 days while

Anand paid 16 for 4 days

Solution – Consider the fixed charge as x and extra charge as y , then

Charge for Latika –

₹ 22 = x + (days–2) y

₹ 22 = x + (6 – 2) y = x + 4y ……..eq(i)

Charge for Anand –

₹16 = x + (total days – 2) y

= x + (4 – 2) y

₹16= x + 2 y ………….eq (ii)

Subtracting eq (ii) from eq (i) ,

x + 4y – (x + 2y) = 22 – 16 …….[In ₹]

$\cancel{x}+4y-\cancel{x}-2y=6$ ………[In ₹]

4y – 2y = 6

2y = 6

y = $\frac{6}{2}$

y = ₹3

Substituting value of y in eq (ii)

₹16 = x + 2 (₹3)

₹16 = x + ₹6

₹16 – ₹ 6 = x

₹10 = x

∴ The fixed charge for first two days is ₹ 10 and additional charges for next days is ₹3 per day.

<sol_end>

<sol_start id=29>

29. Given – Points A(–5,7) , B(–4,–5) , C(–1,–6) and D(4,5) forms quadrilateral

To find – Area enclosed by quadrilateral ABCD

Solution –Dividing the quadrilateral ABCD into ΔABC and ΔADC and finding their areas by

Area of triangle = $\frac{1}{2}$ $\frac{1}{2}\left|x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)\right|$

In ΔABC , x_1 = –5 , x_2 = –4 , x_3 = –1

y_1 = 7 , y_2 = –5 , y_3 = –6

Putting values in the formula,

$= \frac{1}{2}\left|-5\left[-5+(+6)\right]+(-4)(-6-7)+(-1)\left[7-(-5)\right]\right|$

$= \frac{1}{2}\left|-5+52-12\right|$

$= \frac{1}{2}\left|-5+40\right|$

$= \frac{1}{2}\times 35$

$= \frac{35}{2}$ square units

In ΔADC, $x_1 = -5$, $x_2 = 4$, $x_3 = -1$

$y_1 = 7$, $y_2 = 5$, $y_3 = -6$

Area of ΔADC =

$= \frac{1}{2}\left|-5(5+6)+4(-6-7)-1(7-5)\right|$

$= \frac{1}{2}\left|-5(11)+4(-13)-1(2)\right|$

$= \frac{1}{2}\left|-55-52-2\right|$

$= \frac{1}{2}\left|-109\right|$

$= \frac{109}{2}$ square units

Area of quadrilateral ABCD –

= Area of ΔABC + Area of ΔADC

$= \frac{35}{2}+\frac{109}{2}$

= 72 square units

$\therefore$ Area of quadrilateral ABCD is 72 Sq. units

<sol_end>

<sol_start id=30>

30. Given – Two numbers 861 and 1353

To Find – Their HCF using Euclid's algorithm

Solution – Since 1353 > 861 we will apply lemma on 135 and 861

1353 = 861 × 1 + 492

861 = 492 × 1 + 369

492 = 369 × 1 + 123

369 = 123 × 3 + 0

Since remainder is zero when 123 came , 123 is the HCF

∴ HCF of 1353 and 861 is 123,

<sol_end>

<sol_start id=31>

31. Given – Point A(5,–2) and B(–3,2)

To Find – Point on x – axis equidistant force both points.

Solution – Let the point on x-axis be P(x,0)

Using distance formula , $\sqrt{\left(x_1-x_2\right)^2+\left(y_2-x_1\right)^2}$

PA = PB

$$\sqrt{(x-5)^2+(0-(2))^2}=\sqrt{[x-(3)]^2+(0-2)^2}$$

Squaring both sides , we get

$$(x-5)^2+(2)^2=(x+3)^2+(-2)^2$$

$$\cancel{x^2}+25-10x+\cancel{4}=\cancel{x^2}+9+6x+\cancel{4}$$

25 – 9 = 10x + 6x

16 = 16x

x = $\frac{16}{16}=1$

x = 1

∴ The point on the x-axis which is equidistant from points A and B is P(1,0)

<sol_end>

<sol_start id=32>

32. Given – tan A + sin A = m; tan A – sin A = n

To prove – $m^2 - n^2 = 4\sqrt{mn}$

Solution – first solving LHS,

LHS = $(\tan A + \sin A)^2 - (\tan A - \sin A)^2$

$= \tan^2 A + \sin^2 A + 2\tan A\sin A - (\tan^2 A + \sin^2 A - 2\tan A\sin A)$

$= \cancel{\tan^2 A} + \cancel{\sin^2 A} + 2\tan A\sin A - \cancel{\tan^2 A} - \cancel{\sin^2 A} + 2\tan A\sin A$

= 4 tanAsinA

New solving RHS, we get ,

$= 4\sqrt{mn}$

$= 4\sqrt{(\tan A + \sin A)(\tan A - \sin A)}$

$= 4\sqrt{\tan^2 A - \sin^2 A}$

$= 4\sqrt{\frac{\sin^2 A - \sin^2 A}{\cos^2 A}}$ …….($\tan^2 A - \sin^2 A\cos^2 A$)

$= 4\sqrt{\frac{\sin^2 A - \cos^2 A\sin^2 A}{\cos^2 A}}$

$= 4\sqrt{\frac{\sin^2 A(1 - \cos^2 A)}{\cos^2 A}}$

$= 4\sqrt{\frac{\sin^{\cancel{2}} A \times \sin^{\cancel{2}} A}{\cos^{\cancel{2}} A}}$ …….[$\sin^2 A = 1 - \cos^2 A$]

$= \frac{4\sin A \times \sin A}{\cos A}$

= 4 tanAsinA

∴ LHS = RHS, Hence proved

<sol_end>

<sol_start id=33>

33. Given – liner equations x–y+1=0 , 3x + 2y + 2 = 0

In x – y + 1 = 0 , when

x	1	2	5
y	2	3	6

In 3x + 2y – 23 = 0

x	1	2

y 4.5 3

<sol_end>

<sol_start id=34>

Q.34 Given – $p(x) = 2x^3 - x^2 - 4x + 2$ with two

Zeroes – $\sqrt{2}$ and $-\sqrt{2}$

To find – third zero of p(x)

Solution – In p(x) , a = 2 , b = –1 , c = – 4 , d = 2

Using , a + B + r = $\frac{-b}{a}$

Substituting values of a, B and b and a in the formula

$$\left(\sqrt{2}\right)+\left(-\sqrt{2}\right)+r=\frac{+(+1)}{2}$$

O + r = $\frac{1}{2}$

r = $\frac{1}{2}$

$\therefore$ The third root of the polynomial is $\frac{1}{2}$.

<sol_end>

<sol_start id=35>

35. Given – line segment with end points (4, –3) and (8, 5) divided internally in ration 3:1

To Find – Coordinates of the point of division

Solution – Consider the point as P ,

Using the section o=formula to find ,

P = $\left(\frac{x_2m_1+x_1m_2}{m_1+m_2},\frac{y_2m_1+y_1m_2}{m_1+m_2}\right)$

Here $x_1 = 4$, $x_2 = 8$, $m_1 = 3$, $m_2 = 1$

$y_1 = -3$, $y_2 = 5$

Putting values ,

$$P = \left(\frac{8(3)+4(1)}{3+1}, \frac{5(3)+(-3)(1)}{3+1} \right)$$

$$P = \left(\frac{24+4}{4}, \frac{15-3}{4} \right)$$

$$P = \left(\frac{28}{4}, \frac{12}{4} \right)$$

P = (7 , 3)

∴ The point P (7, 3) divide the line joining (4,–3) and (8,5) into 3 : 1

<sol_end>

<sol_start id=36>

<sub_id = 1>

36. (i) → (2)(0, 12)

<sub_id end>

<sub_id = 2>

(ii) → (1) (3,5)

<sub_id end>

<sub_id = 3>

(iii) → (3) 5 m

<sub_id end>

<sol_end>

<sol_start id=37>

<sub_id = 1>

37. (i) → (3) Parabola

<sub_id end>

<sub_id = 2>

(ii) → (3) 0

<sub_id end>

<sub_id = 3>

(iii) → (1) 3, 2

<sub_id end>

<sol_end>

<sol_start id=38>

<sub_id = 1>

38. (i) → (3) 24 cm^2

<sub_id end>

<sub_id = 2>

(ii) →

<sub_id end>

<sub_id = 3>

(iii) → (2) 4:9

Or

<sub_id = 3>

(iii) → (4) Converse of Pythagoras theorem

<sub_id end>

<sol_end>